

Scalability & Future Plan

Date	06 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Application currently runs on local system	Limited accessibility for multiple users	High
2	Limited language support for voice analysis	Users can mainly analyze English explanations	Medium
3	No permanent user history storage	Previous analysis reports are not saved in database	Medium

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports single/local users through Streamlit	Deploy on cloud platform to support multiple users
2	Data Storage	Uses local storage for audio files and reports	Integrate database storage like MongoDB/PostgreSQL
3	Performance	AI models execute on local CPU environment	Use cloud GPU services for faster AI processing
4	Security	Basic application-level handling	Add authentication and secure user data management

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Real-time voice analysis and live feedback system	3 Months	Provides instant evaluation while users speak
Phase 3	Multilingual support and user account management	6 Months	Allows more users and saves individual progress
Phase 4	Advanced AI learning recommendations and cloud deployment	1 Year	Creates a scalable personalized learning platform